



TECHNICAL NOTICE

Third-Party Notices

Licenses, provenance, redistribution obligations, and verified component identities.

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VirtualGlove's original source, documentation, and project artwork are licensed under the repository's MIT License. That license does not replace the terms that apply to third-party software, models, firmware components, or tools.

This document records what a release includes, what an installer obtains from the host system, and the source and license obligations for the modified Nestopia core. VirtualGlove does not distribute ROM images, Nintendo software, scanned game artwork, or publisher assets.

Release contents at a glance

Component	How VirtualGlove uses it	Distribution and license
MediaPipe 0.10.35	Hand landmark recognition on the Controller	Modified ARM64/Python 3.12 wheel included; Apache License 2.0
Google Hand Landmarker	Model used by MediaPipe	Unmodified model included; Apache License 2.0
Arduino Zephyr loader and flash configuration	Loads Matrix firmware on Arduino UNO Q	Two unmodified platform files included; Apache License 2.0
Modified Nestopia libretro core	Native Super Glove Ball input	GPLv2 binary and exact corresponding source included for RetroPie, Recalbox, Batocera, and LaunchBox
NumPy and headless OpenCV	Numerical and camera support	Installed as Python dependencies; BSD-3-Clause and Apache License 2.0 respectively
evdev	Linux console input support	Installed where required; BSD-3-Clause
Python Cryptography	Authenticated LaunchBox communication	Installed in the isolated LaunchBox runtime; Apache License 2.0 or BSD-3-Clause
uhubctl	Optional USB camera power recovery	Installed from Debian when supported; GPLv2 or later
RetroArch, FCEUmm, stock Nestopia, RetroPie, Recalbox, Batocera, and LaunchBox	Console and emulator environment	Supplied separately by the selected platform; their upstream licenses apply

Keep [LICENSE](#), this notice, [licenses/Apache-2.0.txt](#), and [licenses/GPL-2.0.txt](#) with redistributed VirtualGlove packages. Components installed by a platform or package manager retain their own notices and license files.

Controller recognition components

MediaPipe 0.10.35 ARM64 wheel

The Controller package includes one compiled MediaPipe wheel, avoiding a MediaPipe build on the UNO Q during installation. The filename retains its historical `powerglove.cpu1` build tag because that tag is part of the published Python package identity; it is not a service, application, or user-facing product name.

```
python/worker-wheels/mediapipe-0.10.35+powerglove.cpu1-cp312-cp312-linux_aarch64.whl
```

Property	Value
Component	MediaPipe 0.10.35 for CPython 3.12, Linux ARM64
Upstream project	< https://github.com/google-ai-edge/mediapipe >
Upstream source commit	f8ef212d5c962c0e853db7e59d217056b187084b
License	Apache License 2.0
VirtualGlove packaged-wheel SHA-256	6d29bfc33daebd8e47ff9a75d09ae8c032cdcc74445ba365c5aa78a85a6a2d2e

VirtualGlove modifications

The wheel was built from the identified upstream source for the UNO Q. It retains upstream source headers and MediaPipe's Apache 2.0 license at [mediapipe-0.10.35+powerglove.cpu1.dist-info/licenses/LICENSE](#).

VirtualGlove repackaged it for a headless Controller and rebuilt its [RECORD](#) integrity list after these metadata changes:

- The unused `jax` dependency declaration was removed.
- The unused `jaxlib` dependency declaration was removed.
- Headless OpenCV was pinned to `opencv-contrib-python-headless==4.11.0.86`.

The packaged runtime was verified on ARM64 with MediaPipe `0.10.35+powerglove.cpu1`, OpenCV 4.11.0, and no JAX installation. Production uses the four-thread XNNPACK CPU graph. Experimental GPU paths described in the [Engineering Journey](#) are not selected during gameplay. Do not replace this wheel without repeating dependency, camera, recognition, replay, and thermal validation.

Google Hand Landmarker model

VirtualGlove uses Google's float16 Hand Landmarker task bundle.

Property	Value
VirtualGlove Controller runtime path	<code>data/models/hand_landmarker.task</code>
Official download	< https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/float16/1/hand_landmarker.task >
License	Apache License 2.0
SHA-256	fbc2a30080c3c557093b5ddfc334698132eb341044ccee322ccf8bcf3607cde1
Size	7,819,105 bytes

The unmodified recovery copy is [models/hand_landmarker.task](#), and its digest is recorded in [models/SHA256SUMS](#). Its Apache 2.0 license is in [licenses/Apache-2.0.txt](#). Google's official Hand Landmarker [documentation](#) links a [model card](#) that identifies the Apache License 2.0. No model modifications were made.

When vision first starts, VirtualGlove verifies the bundled model and copies it into private persistent storage. A damaged cache is replaced from the bundle; a damaged bundle is rejected. The pinned official download is used only when no bundled model is available. **Gestures off** does not open or install the model.

Python runtime dependencies

VirtualGlove's package metadata is the authoritative dependency list. These packages are installed from the configured Python package source rather than copied into the repository.

Package	Purpose	Upstream license
NumPy	Numerical arrays used by camera and recognition code	BSD-3-Clause
OpenCV headless	Camera capture and image processing without a desktop GUI	Apache License 2.0
evdev	Linux input-device access used by console receivers	BSD-3-Clause
Cryptography	Authenticated LaunchBox runtime communication	Apache License 2.0 or BSD-3-Clause

Installed package metadata and license files remain authoritative for the exact versions selected by the installer.

Matrix firmware toolchain

The verified Matrix firmware toolchain is pinned in [sketch/sketch.yaml](#):

- Arduino Zephyr platform **1.0.0**
- Arduino_RouterBridge **0.4.3**
- Arduino_RPCLite **0.3.0**
- ArxContainer **0.7.0**
- ArxTypeTraits **0.3.2**
- DebugLog **0.8.4**
- MsgPack **0.4.2**

The build resolved Arduino_LED_Matrix **0.1.3** from that platform. Preserve the dependency pins when synchronizing with App Lab; its shortened generated configuration is not a replacement for the project's complete configuration. Revalidate compilation and device operation before changing a pin.

Ordinary UNO Q release archives redistribute two unmodified artifacts from the Arduino Zephyr platform 1.0.0: the UNO Q loader [zephyr-arduino_uno_q_stm32u585xx.elf](#) and [variants/arduino_uno_q_stm32u585xx/flash_sketch.cfg](#). The latter retains its Arduino copyright and Apache-2.0 SPDX header. The platform source is <<https://github.com/arduino/ArduinoCore-zephyr/tree/1.0.0>>; its Apache 2.0 license is included as [licenses/Apache-2.0.txt](#). The adjacent release manifest records exact sizes and SHA-256 values. VirtualGlove's separately compiled Matrix sketch remains MIT-licensed project code; its linked Arduino libraries retain their upstream licenses.

Redistributed platform artifact	SHA-256
zephyr-arduino_uno_q_stm32u585xx.elf	39d4a4fd47241663323f6e04f94dd8f5a9f9ad6582cf1df37f9709b74026adcd
flash_sketch.cfg	38706cee1f9ff2e53364a47129d1c1aea9bb9687ed26d7d70b4a9f9bc5bca60c

Modified Nestopia libretro core

The optional [lr-nestopia-powerglove](#) core is built from libretro Nestopia revision [5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e](#). Nestopia identifies its license as the GNU General Public License, version 2. VirtualGlove's MIT license does not replace the license of Nestopia or the resulting modified core. The corresponding licence text is distributed as [licenses/GPL-2.0.txt](#).

Property	Value
Component	libretro Nestopia with the VirtualGlove native-state patch
Upstream project	< https://github.com/libretro/nestopia >
Pinned revision	5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e
Upstream license	GNU General Public License, version 2
Local modification	native/nestopia-powerglove/nestopia-powerglove.patch
Patch SHA-256	272222b3f55093e901f64228e2adcd82dbedaba77360cf9bfb4c829b6652f706
Windows portability patch	native/launchbox/nestopia-windows.patch
Windows patch SHA-256	88c7bae02bfdb1cda8866dad676710889da7910a8ac695a2a9eb4b62b72e4e7
Modified upstream files	libretro/libretro.cpp ; source/core/input/NstInpPowerGlove.cpp
Modification ledger	This guide, under Nestopia modification ledger
Build recipe	scripts/build-nestopia-powerglove.sh ; pinned RetroPie cross-build environments recorded in its manifest; Recalbox and Batocera target/matrix wrappers; Windows target wrapper scripts/build-launchbox-nestopia-powerglove.sh
Built filename	nestopia_powerglove_libretro.so or nestopia_powerglove_libretro.dll

Binary and source distribution

For redistribution, each bundled binary is accompanied by its exact complete corresponding source, the local patch, build instructions, upstream notices, and the GPLv2 license. The manifests are the authoritative checksum ledger.

Platform	Packaged targets	Authoritative manifest or recipe
RetroPie	armv6, armv7, 32-bit armv8, aarch64, x86_64	native/retropie/manifest.json
Recalbox 10.1	rpizero2, rpi3, rpi4_64, rpi5_64, rg353x, odroidgo2, x86_64	native/recalbox/manifest.json
Batocera 43.1	bcm2835, bcm2836, bcm2837, bcm2711, bcm2712, x86_64, rk3326, rk3399, rk3568, rk3588, s905, s905gen2, s905gen3, s922x, sm8250	native/batocera/manifest.json
LaunchBox	Windows x86_64	native/launchbox/manifest.json

Recalbox 10.1 [rpizero2](#) and every other listed target has a separate binary and corresponding source record in the manifest.

The verified LaunchBox core has SHA-256

[5828d3885a79dba494dcb772a6693093a069a9ac76e758df5392c3da05834299](#). Its corresponding source archive has SHA-256 [a1c3e574b8b65195dab63d3865bf70d3371a44a059ea7d08ef67b3c6595f5033](#).

RetroPie, Recalbox, and Batocera packages select a build for the detected architecture, verify it against the manifest, and perform a target-side libretro load and identity check. Compatible later Recalbox and Batocera releases may reuse the newest packaged build for the same architecture, but the load check remains the final gate. Failure leaves FCEUmm available.

The optional RetroPie installation verifies the packaged core and matching source archive for the host processor. The installer places [COPYING](#), this notice, and the modification ledger beside the installed binary. The exact physically validated Pi 4 binary remains a separate 32-bit ARMv8 artifact; original Pi 2 hardware receives the ARMv7 build instead. Stock Nestopia is never replaced.

For every additional prebuilt core, produce a separately identified artifact for the exact tested operating-system target and architecture. Accompany each binary with the exact complete corresponding source archive used to build it, the local patch and build instructions, all upstream notices, and the GPLv2 license. A Git commit or patch URL alone is not the project's binary-distribution plan. ROM images are never part of a source or binary core artifact.

Installed locations are platform-specific:

- RetroPie: `/opt/retropie/libretrocores/lr-nestopia-powerglove/`
- Recalbox: `/recalbox/share/system/virtualglove/native/recalbox/TARGET/VERSION/`
- Batocera: `/userdata/system/virtualglove/native/batocera/TARGET/VERSION/`
- LaunchBox: `%LOCALAPPDATA%\VirtualGlove\native\`

Runtime scope

RetroArch loads the custom core only for a game explicitly assigned to the native route. The launch entry supplies a read-only latest-sample file through `VIRTUALGLOVE_NATIVE_STATE`; the default is `/run/virtualglove/native-state`. Invalid, stale, uncalibrated, lost-tracking, or wrong-profile samples are neutralized.

The patch registers a separate VirtualGlove controller, identifies the core as Nestopia PowerGlove where required by the established binary interface, and selects the native peripheral for the isolated core. The confirmed ten-byte packet carries X, Y, Z, open hand, fist, index point, and Start behaviour. Fields that the exact ROM does not use remain neutral. The compatibility boundary in [VirtualGlove Input Modes](#) separates confirmed behaviour from deliberately unmapped fields.

The local patch changes only `libretro/libretro.cpp` and `source/core/input/NstInpPowerGlove.cpp`. SHA-256 values for both pristine upstream files are recorded below. The build applies the patch only after checking out the pinned commit, rejects unrelated source-tree changes, and verifies that Nestopia's original 22-line Power Glove copyright and GPL header remains byte-for-byte identical.

Nestopia modification ledger

This additive record does not replace Nestopia's source-file headers, copyright notices, Git history, or `COPYING` file. At the pinned revision, `NstInpPowerGlove.cpp` begins with Martin Freij's original 2003-2008 copyright and GPL notice. The patch begins after that complete header and leaves it byte-for-byte unchanged. The protected pristine files have these SHA-256 values:

- `libretro/libretro.cpp`:
`3e6356e56d1c25e2266be155ea2a0a3155d971e60eecba946026422957f5a25f`
- `source/core/input/NstInpPowerGlove.cpp`:
`7328ab1cc9cc902ac129d217c4d47faa247d2d6e7d41b8d8ac7eb634c097e7cd`

The patch adds the versioned latest-sample record, coherence and freshness checks, safe-neutral handling, a separately selectable controller, native packet generation, core identity, and callback cleanup. Cabinet validation corrected camera-to-Nestopia Y orientation. It also added open-hand, fist, index-point, and calibrated depth delivery without changing the version-1 record size.

An optional diagnostic build inserts checked trace hooks after the production patch. It records callback, sequence, publication, and consumption timing in a private CSV. The diagnostic path does not alter upstream headers, the native state ABI, or normal production selection.

Signed controller protocol changes do not alter this libretro patch or require a core rebuild unless the native-state ABI itself changes. The confirmed behaviour is documented in [VirtualGlove Input Modes](#). Bytes 7-8 retain Nestopia's fixed `$00` initialization because their gameplay role has not been established.

Future changes must remain in the local patch and be appended here. Do not replace or prepend project ownership over an upstream header. Keep upstream notices and licenses with source and binary distributions. The guarded build stops if the protected Nestopia Power Glove header changes.

System-supplied tools and emulators

uhubctl

The VirtualGlove Controller host installer uses the distribution-provided [uhubctl](#) command to detect and operate genuine USB per-port power switching. The helper never bundles or modifies this utility, and it never forces it to operate on a hub that it does not report as supported.

Property	Value
Component	uhubctl USB hub per-port power control utility
Upstream project	< https://github.com/mvp/uhubctl >
License	GNU General Public License, version 2 or later
Installed by	Debian package manager on the VirtualGlove Controller host
Tested repository candidate	Debian 13 ARM64 uhubctl 2.6.0-1
Distribution boundary	Not copied into VirtualGlove source or release archives

The root helper first asks [uhubctl](#) about the exact allowlisted hub location and camera port without using its force option. Only a positive capability result permits a port cycle. Unsupported hardware retains the project's identity-checked whole-hub driver fallback. Debian remains responsible for the installed binary and accompanying copyright and license files.

Console and emulator dependencies

VirtualGlove uses console-provided emulator and frontend software but does not include those binaries in its installation archives. Installation and updates remain governed by the platform and upstream licenses.

Component	VirtualGlove use	Upstream and license	Distribution boundary
RetroArch	Libretro frontend used to load FCEUmm and Nestopia (VirtualGlove)	RetroArch , GPLv3	Installed on the console; not modified or redistributed by VirtualGlove
FCEUmm	Default NES core for standard D-pad/button mappings and the complete Super Glove Ball fallback	FCEUmm , GPLv2	Stock console core; not modified or redistributed by VirtualGlove

The deterministic direction benchmark separately builds stock FCEUmm revision [236ccdfc911e84c60fea6b9d0699c2d440a8de14](#) in an isolated working directory. That pin makes the benchmark reproducible; it does not replace the user's installed core, install FCEUmm, or make the benchmark binary a release artifact.

Updating third-party components

Before publishing a third-party update:

1. Pin an official source revision or URL and record its license, size, and SHA-256 here or in the platform manifest.
2. Preserve upstream headers, notices, and complete applicable license texts.
3. Record every local modification and rebuild package integrity metadata.
4. Verify that every binary has its exact corresponding source where the license requires it.
5. Run package, installer, runtime, security, and hardware acceptance checks.
6. Update this document and rebuild the published Third-party Notices PDF.

For a MediaPipe wheel or model update, also test offline installation, checksum failure, camera start, recognition, tracking loss, and thermal behaviour. For a Nestopia update, recheck the protected upstream header, native-state ABI, packet behaviour, exact-ROM controls, architecture manifests, target-side load checks, and FCEUmm fallback. Detailed commands are in the [Configuration Reference](#).

Documentation and website assets

This section records provenance for project assets. These files are not runtime dependencies.

Gesture illustrations and Pixel Pal

The gesture sheets under [docs/images/gestures/](#) and the individual gesture illustrations and original Pixel Pal mascot under [docs/images/gestures/v2/](#) were generated with OpenAI's image-generation tool from project-authored prompts. The v2 prompts are preserved in [docs/images/gestures/v2/prompts.json](#), and the original generated contact sheet was used as the style reference.

The index-curl illustration was subsequently redrawn using a user-supplied hand photograph as its pose reference. Only the illustrated glove is included; the reference photograph is not distributed with the project.

Intentional extra-digit variants are retained for Pixel Pal's Extra-Digit Hunt. No game screenshots, scans, box art, characters, publisher logos, or other third-party source images were supplied to the generator.

The repository applies its MIT License to these curated project assets to the extent the project owner has rights in them. Game names and other third-party marks remain the property of their respective owners.

Application screenshots and website icon. Screenshots in [docs/images/](#) are generated from VirtualGlove's real page templates with temporary player state and sample telemetry. Camera regions use an explicit **Camera preview omitted** placeholder; the capture process does not access a live camera, cabinet, pairing secret, or personal settings. These images document the interface rather than proving hardware delivery. The website icon in [assets/virtualglove-icon.png](#) was derived from [assets/virtualglove-logo.png](#) with the same image-generation tool. Favicon and Apple touch icon variants were resized from that artwork; the original logo is unchanged.